Big Bags

1 second, 256 MB

You have N bags (1 <= N <= 100,000), numbered from 1 to N. Bag i is of size S_i. No two bags have the same size. Also, bags are ordered by their sizes, i.e., $S_i < S_j$, for $i < j$. A bag of size S_i can hold weights of at most 10 S_i. You are also given an integer K.

We say that a bag is **big** if it can hold other K bags **with full weights inside**.

Consider an example where N = 5, and the sizes of the bags are in the table below.

i	1	2	3	4	5
S_i	2	3	4	6	10
Can hold up to	20	30	40	60	100

With bag sizes as above we have:

- When K=1, bags 4 is big because it can hold bag 2 with full weights (of 30) inside. In fact, bags 2,3,4, and 5 are big.
- When K=2, bag 3 is not big because you cannot put other two full bags inside it as it can only hold the weight of 40, but bag 5 is big because you can put bag 2 (with weight 30) and bag 4 (with weight 60) inside.
- When K=3, only bag 5 is big because you can put bag 1, 2, and 3 with the total full weight of 20+30+40 = 90 inside.

Your job is to find the number of big bags.

Input

The first line of the input contains integers N and K (2<=N<=100,000; K <= N).

The next N lines contains the size. Line 1 + i, for 1 <= i <= N, contains an integer S_i (1 <= S_i <= 1,000,000,000).

It is guaranteed that test cases worth 80%, N <= 10,000, and the total weight is at most 1,000,000,000. In other cases, the sum of weights can be very large, so you should use variable type `long long` to keep them.

Output

Your program should outputs the number of big bags.

Example 1

Input	Output
5 2 2 3 4 6 10	2

Example 2

Input	Output
5 3 2 3 4 6 10	1